

What we're doing

For MarInFold's current decontaminated AFDB + ESM-Atlas training corpus, how many documents and source tokens would remain after deduplication at several sequence-identity thresholds if we remove a protein only when a retained protein also has the same structure, and how many similar-sequence/different-structure examples does that rule rescue?

Why

The answer will differ sharply by source. ESM-Atlas was already reduced to one representative per 40%-identity Linclust cluster, so within-ESM loss above 40% should be small; most removable redundancy should come from AFDB's multiple selected members and overlap between AFDB and ESM-Atlas. A sequence-only calculation will overstate the removable fraction, but the structurally discordant rescue rate should be small rather than zero. The #292 supplement should have much more removable sequence redundancy because it deliberately adds homologous cluster members; its structurally selected minority should survive a joint rule more often than its quality-fill majority.

Full-corpus sequence ceiling

A full 69,516,181-document Linclust pass followed by direct Smith-Waterman verification and a no-chaining selector removes 5,022,836 documents (7.23%) at 50% identity and 80% coverage of both chains. It removes 587,171 (0.84%) at 70%, 276,929 (0.40%) at 90%, and 176,767 (0.25%) exact duplicates.

At 50%, 667,093 AFDB rows (16.83%) and 4,355,743 ESM rows (6.64%) are removed. The full local AFDB ledger puts its exact loss at 738,657,274 source tokens (16.66%). The ESM token total still needs a source-local scan of all 3,338 shards; no incomplete-shard extrapolation is presented as exact.

What does TM 0.7 change?

In #292's curated, measured AFDB anchor/anchor gallery, 44 pairs clear 50% identity and 80% bidirectional coverage. Requiring both directional TM-scores to clear 0.7 rescues one pair (2.3%). Raising TM to 0.8 rescues 7 (15.9%); at 0.9 it rescues 24 (54.5%). The operating threshold matters much more than the phrase "structure-aware dedup" suggests.

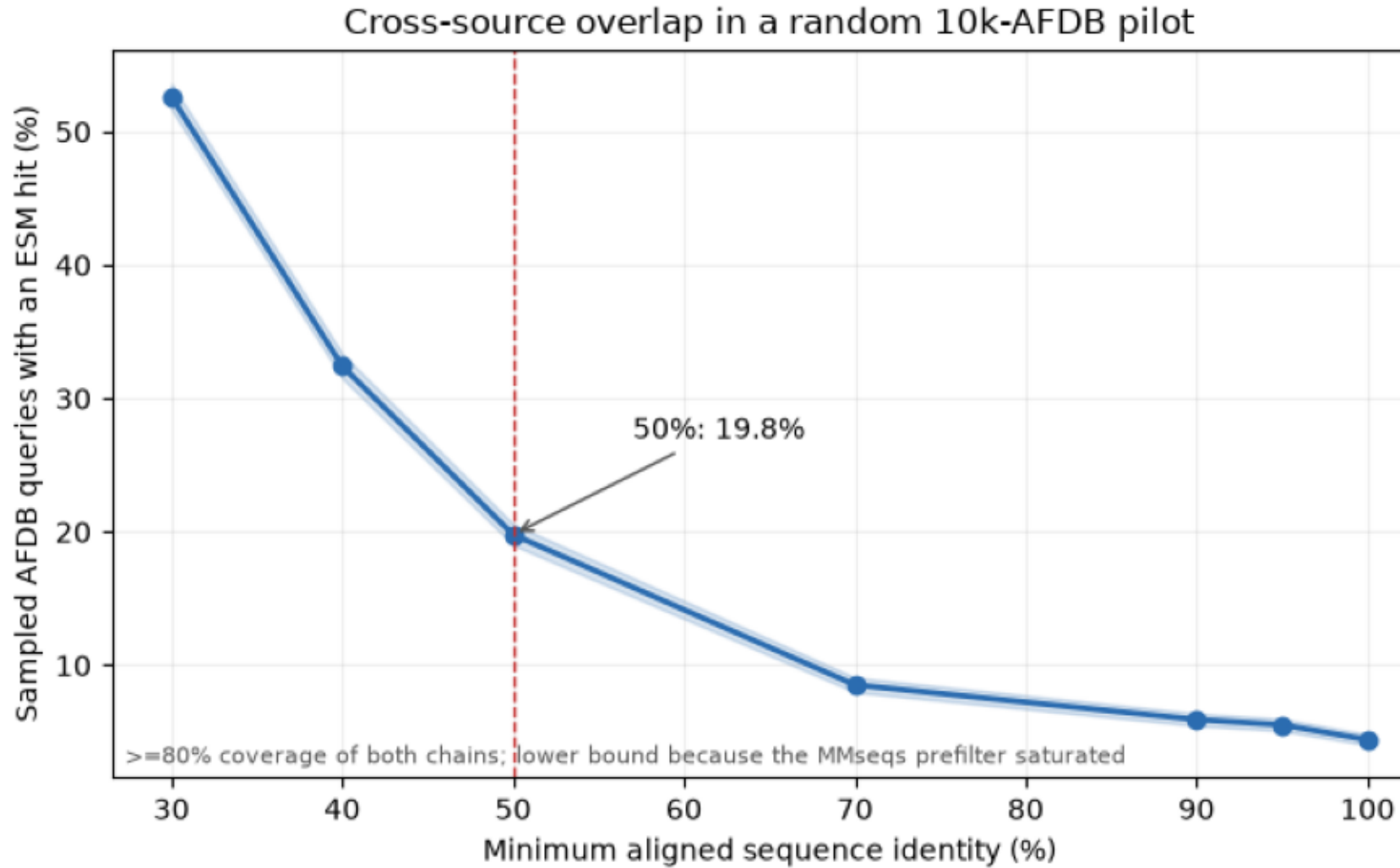
Current interpretation

The 50% sequence-only result is an upper bound on what the requested joint rule may remove: applying $TM \geq 0.7$ can only rescue rows from those 5.02M removals. Within-source ESM redundancy is unexpectedly the largest component (3.75M), showing that one representative per earlier 40%-Linclust cluster was not an all-pairs guarantee. Cross-source witnesses account for 718k removals.

The curated TM gallery suggests a small rescue at 0.7, but it is not a random sample and cannot be multiplied into 5.02M responsibly. The remaining gate is a stratified source-local structure/contact sample, followed by full scoring only if the smoke-run cost is acceptable. No training corpus has been replaced or deleted.

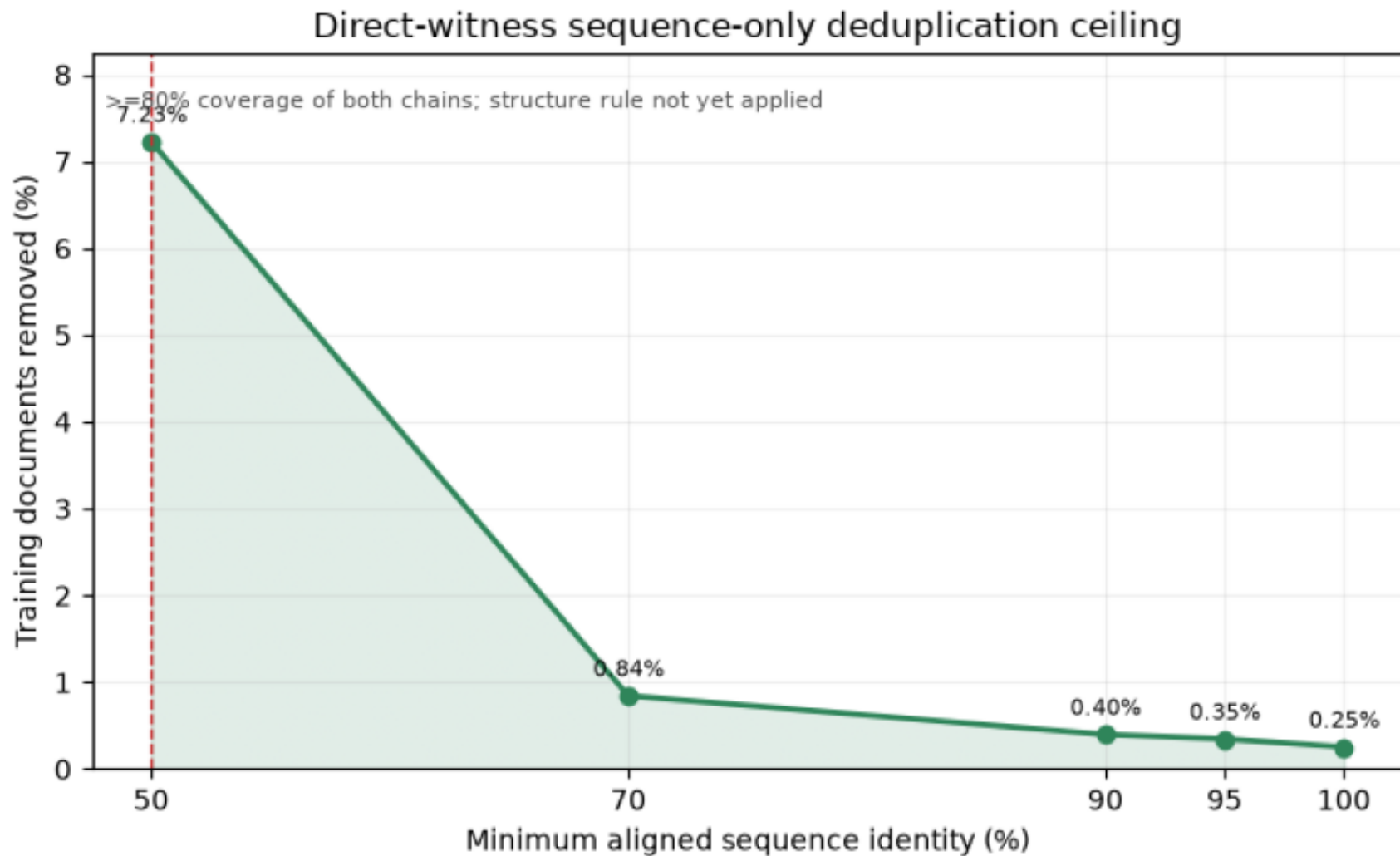
cross_source_sequence_pilot.png

Fraction of 10,000 randomly sampled AFDB rows with at least one current ESM-Atlas match. Counts are lower bounds because the 1,000-candidate prefilter saturated.



direct_sequence_thresholds.png

Documents removed by a deterministic direct-witness selector within frozen 50%-identity Linclust candidate neighborhoods. This is a sequence-only ceiling for joint sequence/structure removal, not the final structure-aware result.



structure_threshold_calibration.png

Share of $\geq 50\%$ -identity, $\geq 80\%$ -bidirectional-coverage measured pairs whose minimum directional TM-score falls below the active threshold.

